

BARTIKA BISWAS

Software Developer

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SUMMARY

Backend Developer with over 1.4 years of experience in building scalable backend systems and AI-driven applications utilizing **Node.js** and **Python**. Proficient in distributed systems, **LLM architectures**, and real-time communication systems with expertise in **Kafka**, **Redis**, and **WebSockets**. Experienced in **AWS**, **Docker**, **Kubernetes**, and performance optimization for high-scale applications.

SKILLS

- Programming Languages:** JavaScript (Node.js), Python, Java
- Frameworks & Libraries:** Express.js, FastAPI, Django, Flask, LangChain, LangGraph, CrewAI, Scikit-learn, TensorFlow, Keras, Transformers
- Databases & Storage:** MongoDB, MySQL, PostgreSQL, Cassandra, Neo4j, Redis, Pinecone, FAISS, AstraDB
- Cloud, DevOps & Version Control:** AWS, Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD, Git, GitHub, Nginx, CDN, S3, EC2, Lambda, CI/CD, SNS, ECR, SQS, Step Functions, RDS
- Backend & System Technologies:** REST APIs, GraphQL, WebSockets, Kafka, Celery, Elasticsearch, Spark, Flink, Microservices Architecture, Distributed Systems, System Design
- AI & LLM Engineering:** RAG Architecture, Prompt Engineering, Multi-Agent Systems, Semantic Search, Advanced Multi-step Reasoning, AI Response Optimization, Retrieval Optimization, transformer-based models, chunk-based pipelines
- Performance & Optimization:** Performance Optimization, Performance Tuning, Load Testing, System Reliability, Database Synchronization, Agile/Scrum Methodologies
- Automation & Workflow:** Data Export Pipelines, Langflow, n8n Automation
- Soft Skills:** Leadership, Public Speaking, Technical Writing, Time Management

WORK EXPERIENCE

OMR India Outsources Pvt. Limited | Backend Developer

Nov 2024 – Present Agra, India

- Developed **scalable backend systems** using Node.js and Python, building **high-performance APIs** and **AI-driven services**.
- Scaled applications from **1 to 1M users** for large file uploads/processing; deployed with integrated cloud services AWS EC2, Docker, Nginx, Kubernetes, and CDN integration.
- Designed and implemented **LLM-based architectures** including LangGraph workflows, **ReAct** agents, **multi-agent systems**, and **advanced multistep reasoning** with RAG pipelines along with **MCP server**
- Contributed to a **Singapore-based production project**, utilizing **Datastax Astra DB**, Langflow, and **n8n automation**
- Implemented **vector database solutions** for **semantic search**, improving **AI response accuracy** and **retrieval efficiency**
- Built **real-time distributed systems** with Kafka, Redis, and WebSockets for **low-latency communication**, integrating AWS SNS/SQS for read/write fan-out
- Performed **load testing** and **performance optimization** using Locust, enhancing **system reliability**
- Developed **tencent-based media transcoding pipelines** for **multi-stage video/image processing** in streaming applications
- Applied **prompt engineering techniques** to enhance **AI chatbot performance**
- Resolved **database synchronization** and **distributed system challenges**, enhancing **system performance**

Sharpenor Tech. | Backend Developer Trainee

Jun 2023 – Aug 2024 Remote

- Built a full-stack **e-commerce platform** with **cart** and **order functionality** using **Node.js**, **Express.js**, and **MySQL**.
- Developed an **expense tracker** with **authentication**, **premium features**, and **AWS S3 integration** for **data export**.
- Created a **group chat application** with **admin roles**, **real-time messaging**, and **media storage** using **AWS S3**.

PROJECTS

AI InnovJAM - AI Assessment Platform

Technologies: FastAPI, transformer-based models, LangGraph, vector databases, Node.js

- Developed an **AI-driven assessment platform** integrating **LangGraph** and **vector databases** for **automated scoring** and **intelligent feedback**, utilizing a **Node.js backend** to support high-concurrency workloads and reduce manual grading by

40%

Reeloid Admin Backend - Streaming Platform

Technologies: Node.js, REST APIs, Redis, Bull, kafka, Elasticsearch,s3, tencent- transcoding, CDN, Firebase

- Developed a **scalable streaming backend**, implementing **REST APIs**, **Redis**, and **Bull** for asynchronous job processing, and **media transcoding pipelines** for video streaming with **ad monetization** and **high-concurrency support** handling over **10,000 uploads**.
 - Optimized and deployed an **LLM model** for **movie recommendation** and **movie script summarization**, enhancing **response accuracy** by **35%** and boosting **recommendation clicks** by **20%**.
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Skillobal - AI Lecture Question Generation System

Technologies: FastAPI, LLM,langchain, chunk-based pipelines, tencent transcoding

- Developed an **AI-powered backend** to convert video content into quizzes and generate summaries by processing **video, audio, and transcripts** into questions.
 - Implemented **chunk-based pipelines** for **audio extraction, transcription, and summarization** to efficiently handle large-scale video datasets.
 - Utilized **advanced prompt engineering** and **asynchronous processing** to reduce manual question creation effort by **40%**.
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On-Screen Marking System - Digital Evaluation Platform

Technologies: schema-based PDF processing, coordinate-based extraction, Poppler, Sharp, websocket, NGINX, Elasticsearch

- Built a backend-driven evaluation system with schema-based PDF processing and coordinate-based extraction, improved image quality, implementing role-based workflows, NGINX load balancing, and Elasticsearch for scalable processing of **10k+** records and fast parallel search

EDUCATION

Bachelor of Technology - Information Technology

University Institute of Technology - Burdwan University, CGPA: 8.6

Aug 2018 - Jul 2022